

1. Given the transfer function

$$H(j\omega) = \frac{j \frac{\omega}{10} (1 + j \frac{\omega}{100})}{(1 + j \frac{\omega}{10}) (1 + j \frac{\omega}{10,000}) (1 + j \frac{\omega}{1,000})}$$

sketch the Bode magnitude and phase plot using the straight-line approximations.

$$Z_1 = 100 \quad +20 \text{ dB/dec}$$

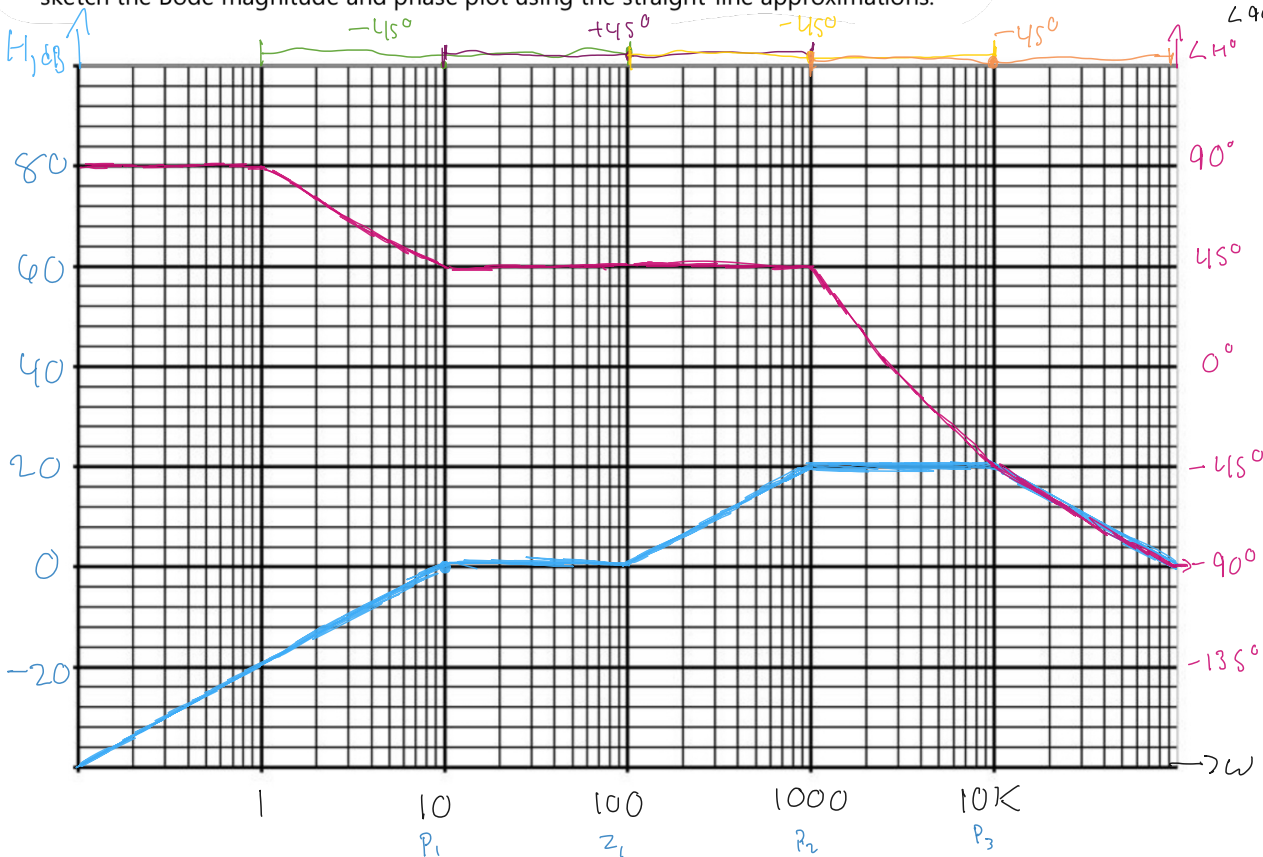
$$P_1 = 10 \quad P_2 = 1k$$

$$P_3 = 10k \quad -20 \text{ dB/dec}$$

$$H(j\omega) \approx j$$

$$|H(j\omega)| \approx j\omega$$

$$\angle 90^\circ$$



2. Given the transfer function

$$H(j\omega) = \frac{60 \cdot j \frac{\omega}{30}}{(1 + j \frac{\omega}{30}) (1 + j \frac{\omega}{20,000})^2}$$

sketch the Bode magnitude and phase plots using straight-line approximations.

slope 20 dB/dec initial

$$P_1 = 30$$

$$P_2 = 20k$$

$$H(j\omega) \approx 2j$$

$$|H(j\omega)| \approx 2$$

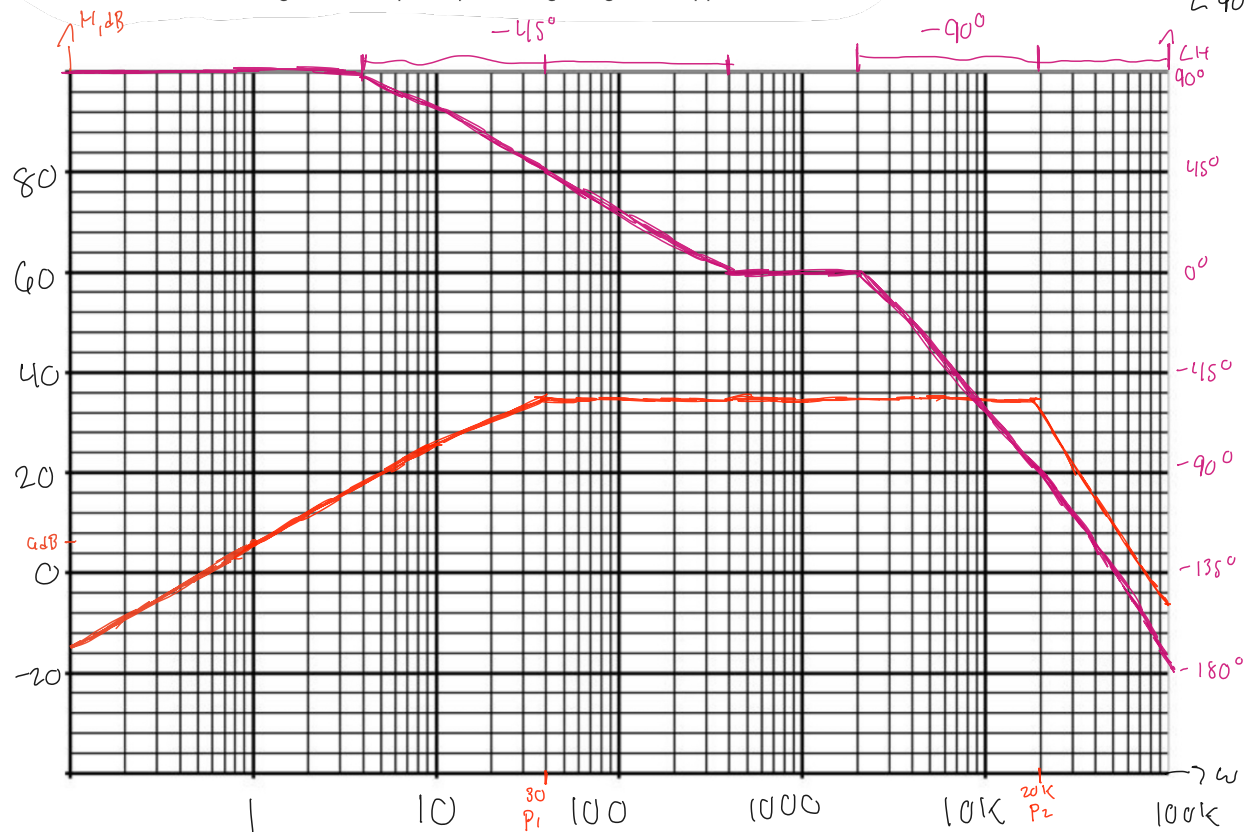
$$20 \log(2) \approx 6 \text{ dB}$$

$$-20 \text{ dB/dec}$$

$$-40 \text{ dB/dec}$$

$$|H(j\omega)| = 2j\omega$$

$$\angle 90^\circ$$

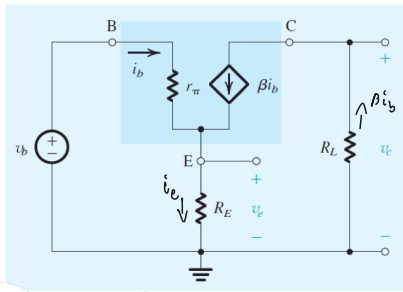


3. For the circuit below show

$$\frac{v_c}{v_b} = \frac{-\beta R_L}{r_{\pi} + (\beta + 1)R_E}$$

and

$$\frac{v_e}{v_b} = \frac{R_E}{R_E + [r_{\pi}/(\beta + 1)]}$$



$$V_c = -R_L \beta i_b$$

$$\frac{V_b - V_e}{r_{\pi}} = i_b$$

$$V_b - V_e = i_b r_{\pi}$$

$$V_b = i_b r_{\pi} + i_b R_E (1 + \beta)$$

$$V_b = i_b (r_{\pi} + R_E (1 + \beta))$$

$$\text{KCL @ E: } i_e = i_b + \beta i_b$$

$$i_e = i_b (1 + \beta)$$

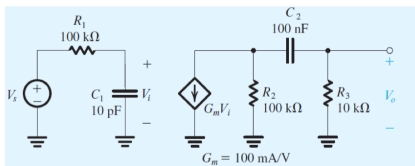
$$V_e = R_E i_e = R_E i_b (1 + \beta)$$

$$\frac{V_c}{V_b} = \frac{-R_L \beta i_b}{i_b (r_{\pi} + R_E (1 + \beta))}$$

$$\frac{V_c}{V_b} = -\frac{R_L \beta}{r_{\pi} + R_E (1 + \beta)}$$

$$\frac{V_e}{V_b} = \frac{R_E i_b (1 + \beta)}{i_b (r_{\pi} + R_E (1 + \beta))} \cdot \frac{1}{(1 + \beta)} = \frac{R_E}{\frac{r_{\pi}}{(1 + \beta)} + R_E}$$

4. For the circuit shown below, first evaluate $H_i(j\omega) = V_i(j\omega)/V_s(j\omega)$ and the corresponding cutoff (corner) frequency. Second, evaluate $H_o(j\omega) = V_o(j\omega)/V_i(j\omega)$ and the corresponding cutoff frequency. Put each of the transfer functions in the standard form discussed in class, and combine them to form the overall transfer function, $H(j\omega) = H_i(j\omega) \times H_o(j\omega)$. Provide a Bode magnitude plot for $|H(j\omega)|$. What is the bandwidth between 3-dB cutoff points?



$$H_i(j\omega) = \frac{V_i(j\omega)}{V_s(j\omega)} \quad \text{low pass}$$

$$V_i = V_s \left(\frac{C_1}{R_1 + C_1} \right)$$

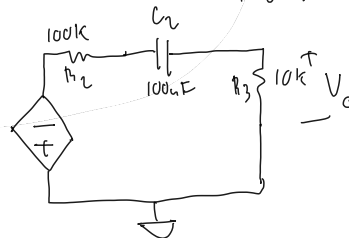
$$\frac{V_i(j\omega)}{V_s(j\omega)} = \frac{1}{1 + j\omega R_1 C_1} = \frac{1}{1 + j\omega \tau_1}$$

$$\tau_1 = R_1 C_1 = 1 \mu$$

$$\omega_1 = \frac{1}{\tau_1} = 1 \cdot 10^6$$

$$f_1 \approx 159 \text{ kHz}$$

$$H_o(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)} \quad \text{high pass}$$



$$V_o = (-100m V_i) \left(\frac{R_2 R_3}{R_2 + R_3 + \frac{1}{j\omega C_2}} \right)$$

$$\frac{V_o}{V_i} = -100m \left(\frac{j\omega R_2 R_3 C_2}{j\omega C_2 (R_2 + R_3) + 1} \right)$$

$$H(j\omega) = H_i(j\omega) H_o(j\omega)$$

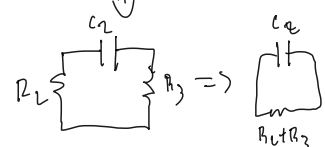
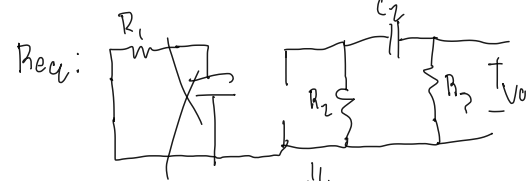
band pass

$$= \left(\frac{1}{1 + j\omega \tau_1} \right) \left(-100m \left(\frac{j\omega R_2 R_3 C_2}{j\omega C_2 (R_2 + R_3) + 1} \right) \right) \cdot \frac{R_2 + R_3}{R_2 + R_3}$$

$$H(j\omega) = \left(\frac{1}{1 + j\omega \tau_1} \right) \left(\frac{-100m R_3}{R_2 + R_3} \left(\frac{j\omega C_2 (R_2 + R_3)}{j\omega C_2 (R_2 + R_3) + 1} \right) \right)$$

$$H(j\omega) = \left(\frac{1}{1 + j\omega \tau_1} \right) (-909 \left(\frac{j\omega \tau_2}{j\omega \tau_2 + 1} \right))$$

$$H(j\omega) = \left(\frac{1}{1 + j\frac{\omega}{\omega_1}} \right) (-909 \left(\frac{j\frac{\omega}{\omega_2}}{1 + j\frac{\omega}{\omega_2}} \right)) = -909 \frac{j\frac{\omega}{40}}{(1 + j\frac{\omega}{1 \cdot 10^6})(1 + j\frac{\omega}{40})}$$



$$\tau_2 = (R_2 + R_3) C_2 = 11m$$

$$\omega_2 = \frac{1}{11m} \approx 90$$

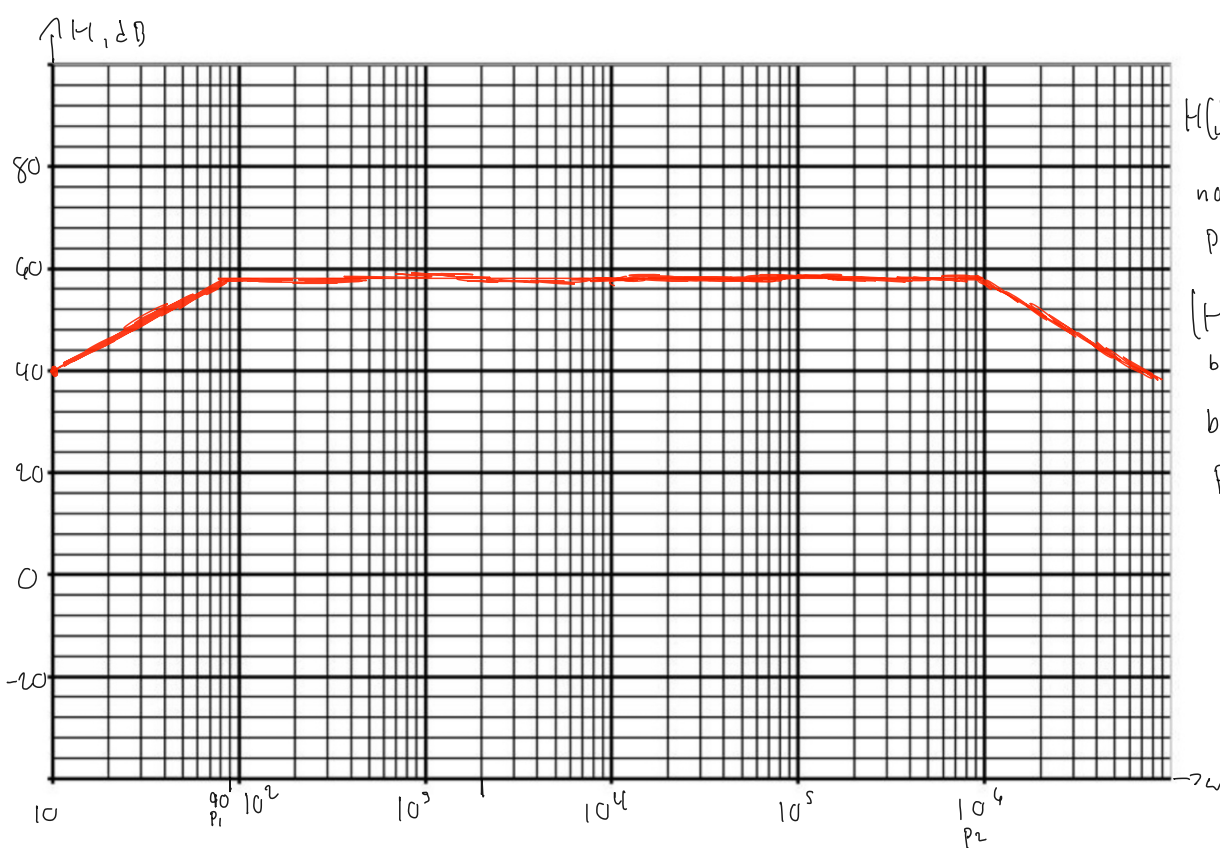
$$f_2 \approx 14.5 \text{ kHz}$$

midband: $\omega_2 \ll \omega \ll \omega_1$

$$\omega_1 \ll 1: \frac{1}{1 + j\omega \tau_1} \approx 1$$

$$\omega_2 \gg 1: -100m \frac{R_2 R_3}{R_2 + R_3 + \frac{1}{j\omega C_2}} \approx -100m \frac{R_2 R_3}{R_2 + R_3} \approx -909$$

since high pass + low pass gives band pass



$$H(j\omega) = -909 \frac{j\omega}{(1+j\frac{\omega}{10^2})(1+j\frac{\omega}{10^4})}$$

no 0's

$$P_1 \approx 90 \quad P_2 \approx 1 \cdot 10^6$$

$$|H(j\omega)| \approx -909 \frac{10}{\omega} \approx 100 = 40 \text{ dB}$$

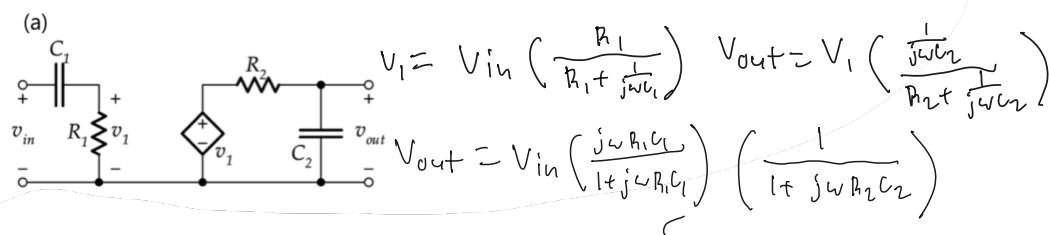
below midband

bandwidth between 3dB cutoff

$$\text{points: } f_1 - f_2 \approx 159 \text{ K} - 14.5$$

$$\text{bandwidth} \approx 159.14 \text{ [KHz]}$$

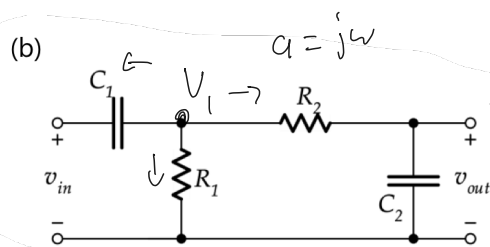
5. Find the transfer function, $\frac{v_{out}}{v_{in}}$, for the following circuits. For (a) and (b), $R_1 = R_2 = 1 \text{ K}\Omega$, $C_1 = 1 \mu\text{F}$, and $C_2 = 100 \mu\text{F}$. For (c), $R_1 = 1 \text{ K}\Omega$, $R_2 = 100 \Omega$, $C_1 = C_2 = 1 \mu\text{F}$, and $K = 60$.



$$\frac{V_{out}}{V_{in}} = \frac{j\omega R_1 C_1}{1 + j\omega R_2 C_2 + j\omega R_1 C_1 - \omega^2 R_1 R_2 C_1 C_2}$$

$$\frac{V_{out}}{V_{in}} = \frac{j\omega R_1 C_1}{j\omega (R_2 C_2 + R_1 C_1) - \omega^2 R_1 R_2 C_1 C_2 + 1}$$

$$\frac{V_{out}}{V_{in}} = \frac{j\omega 0.001}{j\omega (0.101) - \omega^2 (0.0001) + 1}$$



$$V_o = V_1 \frac{\frac{1}{\alpha C_2}}{R_2 + \frac{1}{\alpha C_2}} = V_1 \left(\frac{1}{1 + \alpha R_2 C_2} \right)$$

$$\frac{V_o}{V_{in}} = \frac{\alpha C_1}{\alpha C_1 + \frac{1}{R_1} + \frac{\alpha C_2}{1 + \alpha R_2 C_2}} \cdot V_o = V_T \left(\frac{1}{1 + \alpha R_2 C_2} \right)$$

$$\frac{V_o}{V_{in}} = \frac{\alpha C_1}{\left(\alpha C_1 + \frac{1}{R_1} + \frac{\alpha C_2}{1 + \alpha R_2 C_2} \right) (1 + \alpha R_2 C_2)}$$

$$\frac{V_o}{V_{in}} = \frac{\alpha C_1}{(1 + \alpha R_2 C_2) \left(\alpha C_1 + \frac{1}{R_1} \right) + \alpha C_2}$$

$$\frac{V_o}{V_{in}} = \frac{\alpha C_1}{\alpha C_1 + \frac{1}{R_1} + \alpha^2 R_2 C_1 C_2 + \alpha \frac{R_2 C_2}{R_1} + \alpha C_2}$$

$$\frac{V_o}{V_{in}} = \frac{\alpha R_1 C_1}{\alpha R_1 C_1 + 1 + \alpha^2 R_1 R_2 C_1 C_2 + \alpha R_2 C_2 + \alpha R_1 C_2}$$

$$\frac{V_1 - V_{in}}{\frac{1}{\alpha C_1}} + \frac{V_1}{R_1} + \frac{V_1 - V_o}{R_2} = 0$$

$$\alpha C_1 (V_1 - V_{in}) + \frac{V_1}{R_1} + \frac{V_1}{R_2} \left(1 - \frac{1}{1 + \alpha R_2 C_2} \right) = 0$$

$$\alpha C_1 (V_1 - V_{in}) + \frac{V_1}{R_1} + \frac{V_1}{R_2} \left(\frac{1 + \alpha R_2 C_2 - 1}{1 + \alpha R_2 C_2} \right) = 0$$

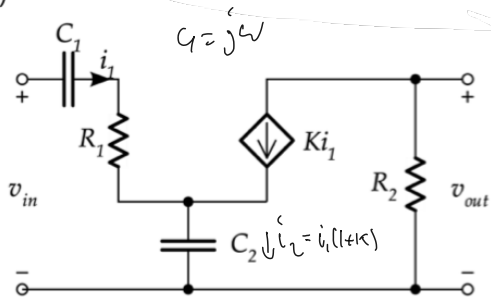
$$\alpha C_1 (V_1 - V_{in}) + \frac{V_1}{R_1} + V_1 \frac{\alpha C_2}{1 + \alpha R_2 C_2} = 0$$

$$V_1 \left(\alpha C_1 + \frac{1}{R_1} + \frac{\alpha C_2}{1 + \alpha R_2 C_2} \right) = \alpha C_1 V_{in}$$

$$\frac{V_1}{V_{in}} = \frac{\alpha C_1}{\alpha C_1 + \frac{1}{R_1} + \frac{\alpha C_2}{1 + \alpha R_2 C_2}}$$

$$\frac{V_o}{V_{in}} = \frac{a R_1 C_1}{a(R_1 C_1 + R_2 C_2 + R_1 C_2) + 1 + a^2 R_1 R_2 C_1 C_2} = \frac{j\omega 0.001}{j\omega(0.201) + 1 + (j\omega)^2 0.0001}$$

(c)



$$V_o = -K i_1 R_2$$

$$V_o = -K \left[\frac{V_{in} a C_1 C_2 R_2}{C_2 + C_1(1+K) + a C_1 C_2 R_1} \right]$$

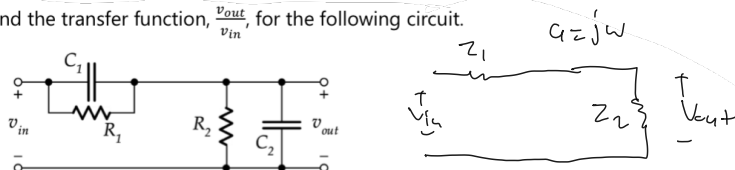
$$\frac{V_o}{V_{in}} = - \frac{j\omega 0.1 \cdot 10^{-9}}{0.2 \cdot 10^{-6} + j\omega 10^{-9}}$$

$$V_{in} = i_1 \left(\frac{1}{a C_1} + R_1 + \frac{1+K}{a C_2} \right)$$

$$V_{in} = i_1 \left(\frac{C_2 + C_1(1+K) + a C_1 C_2 R_1}{a C_1 C_2} \right)$$

$$i_1 = \frac{V_{in} a C_1 C_2}{C_2 + C_1(1+K) + a C_1 C_2 R_1}$$

6. Find the transfer function, $\frac{v_{out}}{v_{in}}$, for the following circuit.



By proper choice of the components C_1 , R_1 , C_2 , and R_2 , the transfer function of this circuit can be made independent of frequency. In that case, the circuit is called a compensated attenuator and can be used as an oscilloscope probe. In this application, the signal from a circuit being viewed on an oscilloscope is attenuated, and the input impedance of the oscilloscope is compensated by the probe so that it does not distort the measurement. Assume that this circuit, R_1 and C_1 represent the probe, while R_2 and C_2 represent the oscilloscope input. For an oscilloscope with an input of $1 \text{ M}\Omega$ and 30 pF , design a 10:1 probe, that is, a probe which attenuates the signal by a factor of 10.

frequency independent; $j\omega$ must cancel
(a)

$$1 + a R_1 C_1 = 1 + \frac{a R_1 R_2 (C_1 + C_2)}{R_1 + R_2}$$

$$R_1 C_1 = \frac{R_1 R_2 (C_1 + C_2)}{R_1 + R_2}$$

$$(R_1 + R_2) C_1 = R_2 (C_1 + C_2)$$

$$R_1 C_1 + R_2 C_1 = R_2 C_1 + R_2 C_2$$

$$R_1 C_1 = R_2 C_2$$

$$Z_1 = \frac{R_1}{a C_1} = \frac{R_1}{1 + a R_1 C_1}$$

$$Z_2 = \frac{R_2}{a C_2} = \frac{R_2}{1 + a R_2 C_2}$$

$$V_o = V_{in} \left(\frac{Z_2}{Z_1 + Z_2} \right)$$

$$= V_{in} \left[\frac{R_2}{(1 + a R_2 C_2) \left(\frac{R_1}{1 + a R_1 C_1} + \frac{R_2}{1 + a R_2 C_2} \right)} \right]$$

$$\frac{V_o}{V_{in}} = \frac{R_2}{(1 + a R_2 C_2) \left(\frac{R_1 + a R_1 R_2 C_2 + R_2 + a R_1 R_2 C_1}{(1 + a R_1 C_1)(1 + a R_2 C_2)} \right)}$$

$$\frac{V_o}{V_{in}} = \frac{R_2 (1 + a R_1 C_1)}{R_1 + R_2 + a R_1 R_2 (C_1 + C_2)}$$

$$\frac{V_o}{V_{in}} = \frac{R_2}{R_1 + R_2} \cdot \frac{1 + a R_1 C_1}{1 + \frac{a R_1 R_2 (C_1 + C_2)}{R_1 + R_2}}$$

but for frequency

independence $\Rightarrow \frac{V_o}{V_{in}} = 0.1 = \frac{R_2}{R_1 + R_2}$
and 10x
attenuation

$$(R_1 + R_2) 0.1 = R_2$$

$$.1 R_1 + .1 R_2 = R_2$$

$$0.1 R_1 = 0.9 R_2$$

$$R_1 = 9 R_2$$

$$R_1 = 9(1 \text{ M}) = 9 \text{ M}\Omega$$

$$\therefore R_1 C_1 = R_2 C_2$$

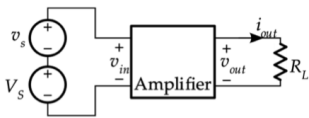
$$C_1 = \frac{1 \text{ M}(30 \text{ pF})}{9 \text{ M}}$$

$$C_1 = 3.33 \text{ pF}$$

7. An amplifier has a transfer characteristic that can be reasonably approximated by the expression

$$v_{OUT} = \begin{cases} -12V & -20V \leq v_{IN} \leq 8V \\ 2v_{IN} - 28V & 8V < v_{IN} \leq 9V \\ 10v_{IN} - 100V & 9V < v_{IN} \leq 11V \\ 2v_{IN} - 12V & 11V < v_{IN} \leq 12V \\ 12V & 12V < v_{IN} \leq 20V \end{cases}$$

- Sketch this transfer characteristic.
- Assume that $V_S = 10.5V$ and $v_s = A \sin(\omega t)$, where $A = 0.3V$ and $\omega = 60 \text{ rad/s}$, as shown in the figure below. For this signal level and DC bias value, what is the voltage gain, A_v ? Assume that $A_v = v_{out}/v_{in}$.
- Assume the above values for V_S and v_s and find an expression for $i_{out}(t)$ when $R = 100\Omega$.
- What are the largest values for v_s that will yield an undistorted output?



b) $V_{in} = V_S + v_s = 10.5 + 0.3 \sin(\omega t)$

$V_{in, \min}; \omega t = \frac{3\pi}{2} \quad 10.5 - 0.3$

$\max; \omega t = \frac{\pi}{2} \quad 10.5 + 0.3$

$10.2 \leq v_{in} \leq 10.8 \quad 9 < v_{in} < 11, \quad v_{out} = 10v_{in} - 100$

$\frac{v_{out}}{v_{in}} = A_v = 10$

d) $v_s = A \sin(60t)$

undistorted $9 \leq v_{in} \leq 11$

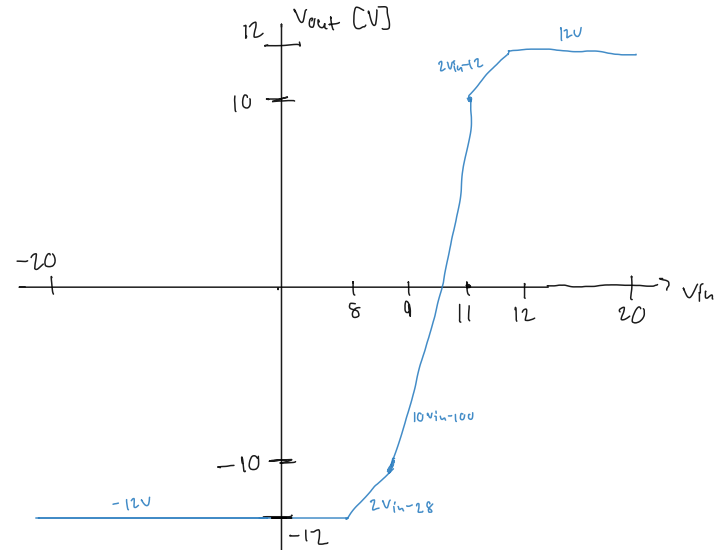
$9 \leq 10.5 + A \sin(60t) \leq 11$

largest v_s value: $10.5 + A \sin(60t) \leq 11$

$A \sin(60t) \leq 0.5$

$\therefore$ largest v_s value when $A = 0.5$ to be undistorted

a)



c) $v_{out}(t) = i_{out}(t) \cdot R$

$10(10.5 + 0.3 \sin(60t)) - 100 = i_{out}(t) \cdot R$

$\frac{5 + 3 \sin(60t)}{100} = i_{out}(t)$

$i_{out}(t) = 50 + 30 \sin(60t) \text{ [mA]}$